Optimizing Wastewater Pumping

Challenge for Junction 2025

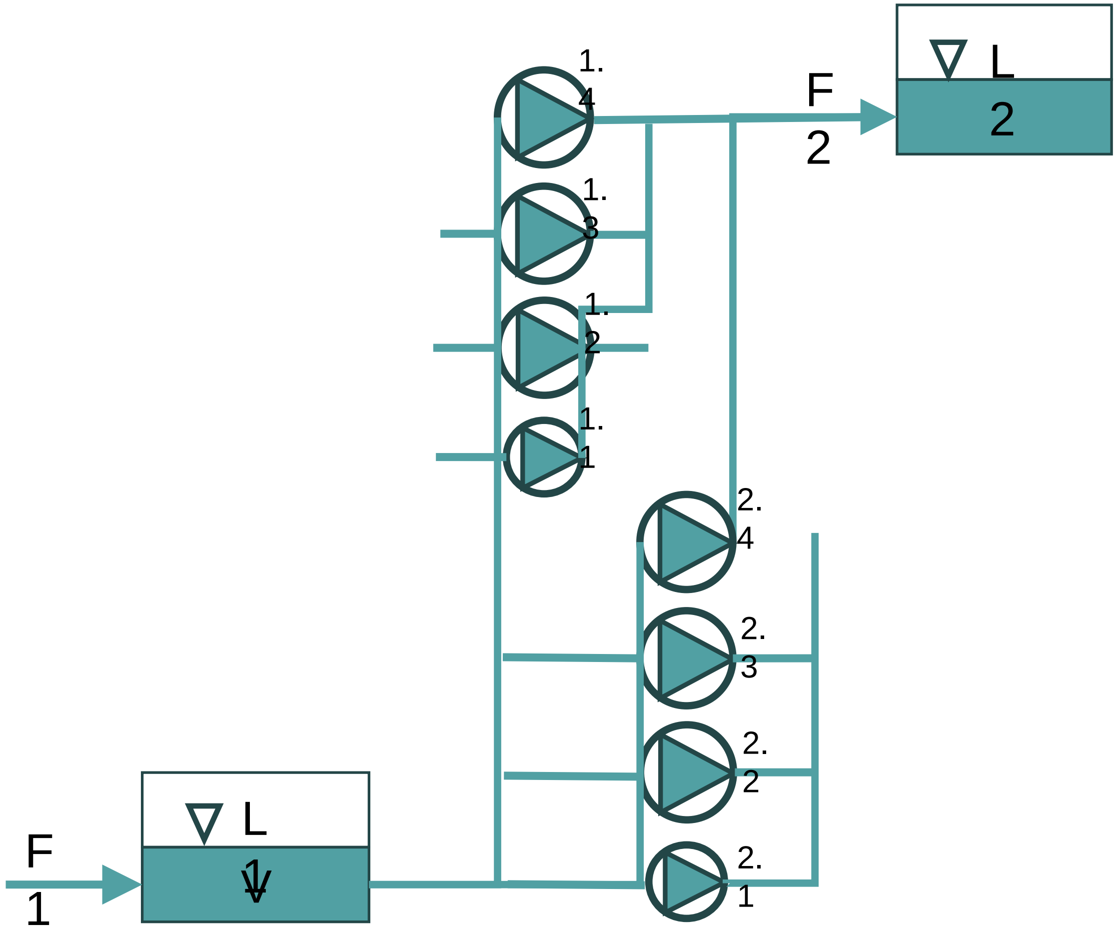
HSY

Task

Design a pump control scheme that minimizes energy costs while fulfilling certain boundary conditions

System Description

- Wastewater is pumped from a collection tunnel to the head of the wastewater treatment plant (WWTP)
- There are 6 big and 2 small pumps
- Cost of pumping is determined by energy consumption and price of electricity
- Energy requirement is determined by the
  - pumped flowrate
  - difference in elevation H (L2 – L1, m)
  - Pump efficiency η, %
- Water level L2 at the WWTP (pump pressure side) is constant
- Water level L1 in the tunnel (pump suction side) is determined by inflow from sewer network F1 and the flow pumped to the WWTP F2



Data

- Data provided from a period of 14 days, dt 15 min
  - Inflow to tunnel F1
  - Suction level L1
  - Volume of water in tunnel V
  - Pumped flow per pump
  - Power intake per pump
  - Drive frequency per pump
  - Electricity price as two alternative time series
- Other data
  - Formula for calculating V as function of L1
  - Pump curves showing dependency of pumped flow, H and η

Boundary conditions

- Pump capacities (see pump curves and data)
- Min and max limits for L1
  - L1min: **0 m**
  - L1max: **8 m**
- L2: **30 m**
- Pumped flow to WWTP (F2) should be as close to constant as possible, with smooth changes
- Pumping must not be stopped entirely
- Pumps must not start and stop frequently (<2h)
- Pumps to be operated at or close to full speed (>47,5 Hz)
- Good pump efficiency is desirable
- Tunnel must be emptied (L1 < 0,5 m) once every day during low inflow
  - Not necessary during rain events (high inflow)

Phenomena & Dynamics

- Large tunnel volume = delay between pump activation and impact
- Seasonality, daily usage cycle: morning & evening peaks, low night consumption
- Storm rain effect: rapid, episodic increase in inflow

Technical Setup & Limitations

- Dataset: 14 d
- Data format: CSV / Excel
- Simulation: Offline Python-based model
- Optional: Feedforward control or multi-agent AI-based logic
- Visualization: Cross-sectional view with topography, tanks, and pumps

Targeted Outcome

- Data-driven, intelligent pump control algorithm
- Accounts for delay, demand peaks, rainfall, and electricity price
- Accounting for pump efficiency is a plus
- Allows scenario testing in a simulation environment

Further Development Potential

- Generic solution applicable to other waste water plants e.g. Viikinmäki (HSY), Sulkavuori (Tampereen keskuspuhdistamo), Kakola (Turun vesi) and others in Finland / globally

- Potential for productization through collaboration

- Possible integration with OPC UA/Digital Twin architecture